

RUNNER-4: Technical Engineering Report

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1. Mission & Mechanical Design

RUNNER-4 is a lightweight, 550g, 12-DOF autonomous quadruped powered by an STM32F411CEU6 microcontroller. Its mechanical framework is engineered for dynamic locomotion, allowing it to traverse complex terrain and navigate tight spaces effectively.

The system incorporates 12 degrees of freedom, utilizing three active revolute joints per leg (coxa, femur, and tibia). This kinematic arrangement provides redundancy in task-space trajectory planning, enabling the controller to decouple body position from orientation and optimize joint velocity distribution.

This adaptability allows the robot to dynamically shift its static margin of stability over the support polygon. Consequently, the system maintains equilibrium across non-planar or irregular surfaces, reducing foot slip and ensuring consistent traction during complex maneuvers.

Mission Subtasks

- **Grid Navigation:** Autonomous traversal and ball retrieval using a 2-DOF robotic arm with a custom end-effector.
- **Wall-Following:** Perimeter boundary navigation featuring real-time ToF gap rejection.
- **Obstacle Clearing:** Passage management using a front-mounted, passive spring-loaded bumper.
- **Color-Based Sorting:** Precise identification and sorting managed via an integrated RGBC sensor assembly.

Structural Specifications

The chassis utilizes a symmetrical 12-DOF articulation layout within a 160×140mm footprint. Constructed from PETG, the 2-tier design prioritizes torsional rigidity and impact resistance. PETG was selected for its superior inter-layer adhesion and fatigue resistance under cyclic loading, ensuring that the structure absorbs high-frequency mechanical shocks without degradation. A bottom-mounted 3S LiPo battery pack keeps the center of mass low, optimizing stability during dynamic gaits.

2. Electronics & Sensors

The processing architecture centers on the STM32F411CEU6, an ARM Cortex-M4 microcontroller running at 100MHz. The integrated Floating-Point Unit (FPU) is critical for real-time inverse kinematics (IK), enabling single-cycle hardware execution of trigonometric operations required for 3-DOF joint angle solutions without the latency of software approximations.

Power distribution utilizes a 3S LiPo battery. A high-current 5V/15A BEC is dedicated to the high-torque servo motors, while a low-noise 3.3V LDO regulator powers the digital logic and sensing peripherals.

To preserve signal integrity, the system employs complete galvanic isolation between the actuator power domain and the digital logic rail. Star-ground topology and separate power planes prevent ground loops and inductive voltage spikes, ensuring that the heavy current draw of micro-servos does not induce logic brownouts or sensor noise.

Sensor Suite

Sensor	Qty	Function
8-ch TCRT5000 IR	1	Line/junction detection
TCS34725 RGBC	1	Ball/floor color classification
VL53L0X ToF	3	Wall/obstacle distance sensing
MPU6050 IMU	1	Body stabilization/tilt detection
Start Button	1	Autonomous start trigger

3. Control & Software

The system operates on a deterministic 50Hz C-firmware control loop, ensuring consistent updates for trajectory generation and motor command outputs.

Core Algorithms

- **IK/Gait:** An analytical 3-DOF IK solver coupled with a 2.0s Bezier-curve trajectory generator. By guaranteeing C2 continuity across trajectories, the system eliminates infinite jerk spikes, reducing mechanical vibration and stress on servo gear trains.
- **Stabilization:** Active body attitude adjustment using an MPU6050 IMU with a complementary filter ($\alpha=0.98$) for real-time pitch and roll correction.
- **Navigation:** A closed-loop PD system utilizes feedback from the IR line array and ToF rangefinders. It incorporates a dynamic gap rejection filter that enforces temporal thresholding—requiring consecutive valid samples before registering a gap—to prevent misinterpreting sensor noise or specular reflections as open pathways.
- **Logic Control:** An 18-state Hierarchical Finite State Machine (HFSM) manages mission sequencing. It includes robust error-handling, such as automated soft bus resets for I2C peripherals, ensuring continuous operation even if individual sensors experience transient timeouts.

Control Implementation

Algorithm	Implementation
3-DOF IK	Trigonometric solver
Crawl Gait	Bezier-curve (2s cycle)
IMU Stabilization	Complementary Filter ($\alpha=0.98$)
Line/Wall Following	PD Controller
Obstacle Pushing	ToF-triggered stance
Mission State	18-state HFSM

4. Calibration, Risk Management & Conclusion

Pre-Flight Calibration Procedure

1. **Inertial Baseline:** Stationary IMU zero-offset correction to nullify gyroscope and accelerometer biases.

2. **Reflectance Normalization:** Ambient IR calibration across dark and light surfaces to set dynamic detection thresholds.
3. **Sensor Profiling:** ToF distance baseline verification and RGBC color balance profiling.
4. **Mechanical Alignment:** Servo neutral stance positioning to eliminate assembly offsets.

Risk Mitigation

We employ active thermal management via custom heatsinks and ventilation to maintain voltage regulator headroom. To prevent hardware damage, the software monitors actuator current; exceeding thresholds triggers un-weighted leg re-positioning to clear obstructions. Furthermore, continuous sensor drift correction for the IMU is performed during every stationary stance phase.

Strategic Overview

Decision	Choice	Rationale
Gait	Crawl	Torque/Stability
Perception	TCRT/TCS	Low-latency
Pushing	Passive	Mechanical reliability
Battery	2S LiPo	Optimized weight/runtime

Team Delegation

- **Mechanical:** CAD, assembly, and bumper fabrication.
- **Electronics:** Power distribution and wiring architecture.
- **Firmware:** IK solvers, sensor drivers, and HFSM logic.
- **Integration:** Calibration, testing, and competition readiness.

Conclusion

With a 37-minute runtime and an optimized mass of ~550g, RUNNER-4 is engineered for maximum reliability. The combination of static crawl stability and robust state-machine logic ensures consistent performance within the competition environment.